

Multiple Choice (10 marks)

1. Of the following ions, which has the smallest radius?
 - (a) K^+
 - (b) Ca^{2+}
 - (c) Sc^{3+}
 - (d) Rb^+
2. Which of the following is the most common naturally-occurring form in which silicon is found?
 - (a) Metallic element
 - (b) Sulfide
 - (c) Fluoride
 - (d) Oxide
3. A set of hybrid sp^3 orbitals for a carbon atom is given above. Which of the following is NOT true about the orbitals?
 - (a) The orbitals are degenerate.
 - (b) The set of orbitals has a tetrahedral geometry.
 - (c) These orbitals are constructed from a linear combination of atomic orbitals.
 - (d) Each hybrid orbital may hold four electrons.
4. Elements with partially filled 4 f or 5 f orbitals include all of the following EXCEPT
 - (a) Cu
 - (b) Gd
 - (c) Eu
 - (d) Am
5. Which of the following experimental observations were explained by Planck's quantum theory?
 - (a) Blackbody radiation curves
 - (b) Emission spectra of diatomic molecules
 - (c) Electron diffraction patterns
 - (d) Temperature dependence of reaction rates

True / False (10 marks)

Answer True or False

6. True or False: The magnetic field responsible for the polarization of 2.5×10^{-6} for ^{13}C at 298 K is 2.9 T.
6. True or False: Thionyl chloride, SOCl_2 , has a molecular geometry that is best described as tetrahedral, which is common for molecules with four regions of electron density.
7. True or False: All molecular bonds absorb IR radiation equally, regardless of their type or functional group.
8. True or False: Ammonia (NH_3) is the strongest base in liquid ammonia, surpassing all other species.
9. True or False: The ^{13}C NMR spectrum of 2-methylpentane exhibits five distinct chemical shifts due to its unique structural isomers.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process by which cobalt-60 is produced from cobalt-59, including the role of neutrons in this transformation and its implications for radiation therapy in cancer treatment.
12. Discuss the EPR spectrum characteristics of a rigid nitronyl nitroxide diradical, particularly focusing on the number of lines produced when $J \ll a$, and explain the underlying principles involved.

End of Paper